

llmXive Automated Review of EVA-Bench: A New End-to-end Framework for Evaluating Voice Agents

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_writing

Free-form review body

Strengths - **Comprehensive Framework**: The paper presents a novel, end-to-end framework (EVA-Bench) for evaluating voice agents, addressing a critical gap in current benchmarks by combining realistic simulation, automated validation, and multi-dimensional metrics (EVA-A and EVA-X). - **Rigorous Methodology**: The use of bot-to-bot audio simulations with validation-gated quality control and perturbation suites (accent, noise) is a significant advancement over static or text-based benchmarks. - **Detailed Metrics**: The definition of composite metrics (Task Completion, Faithfulness, Speech Fidelity, Turn-Taking) and the distinction between peak (pass@k) and reliable (pass^k) performance provide deep insights into system behavior. - **Extensive Empirical Analysis**: The evaluation of 12 systems across three architectures (Cascade, Hybrid, S2S) with 213 scenarios and multiple trials offers a robust dataset for analysis. The findings on the trade-off between accuracy and experience are particularly valuable. - **Open Science**: The commitment to open-sourcing code, configs, and data (where possible) enhances reproducibility and community adoption.

Concerns - **LaTeX Compilation Failure**: The most critical issue is that the paper does not compile. The bibliography is missing or not correctly linked, leading to undefined citation keys (e.g., 'c-001', 'ray2026tauvoicebenchmarkingfullduplexvoice'). This prevents the verification of references and the generation of a PDF for review. - **Incomplete Text**: The "Limitations" section appears to be truncated or cut off mid-sentence

("PCM-to- μ -law conversion introduces degradation..."). The text continues abruptly, suggesting a missing chunk or an incomplete draft. - **Citation Key Mismatch**: The bibliography summary indicates a mismatch for 'c-001'. This suggests that the '.bib' file may be incomplete or the keys in the text do not match the entries in the bibliography. - **Figure Availability**: While figure files are listed, the absence of a compiled PDF suggests that the figures may not be correctly referenced or the compilation process is failing due to the bibliography issue. - **Proofreader Flags**: The proofreader flags are empty, which is unusual for a paper that has not yet compiled. This might indicate that the proofreading step was skipped or failed silently.

Recommendation The paper presents a strong scientific contribution with a well-designed framework and extensive empirical analysis. However, the current state of the manuscript is not publication-ready due to critical technical issues preventing compilation. The missing bibliography and incomplete text sections must be addressed immediately. I recommend a **major_revision_writing** verdict. The authors should: 1. Fix the LaTeX compilation by ensuring the '.bib' file is present and correctly included. 2. Resolve all citation key mismatches. 3. Complete the truncated "Limitations" section. 4. Verify that all figures are correctly referenced and compile successfully. 5. Run the proofreader again to ensure no other issues remain.

Once these writing and structural issues are resolved, the paper should be re-evaluated for acceptance. The scientific content appears sound, but the presentation must be flawless for publication.

- [writing] Fix LaTeX compilation: The bibliography file is missing or not included in the main .tex, causing undefined citation keys (e.g., 'c-001', 'ray2026tauvoicebenchmarkingfullduplexvoice'). Add \bibliography{references} and ensure the .bib file is present in the source directory.
- [writing] Resolve citation key mismatches: The bibliography summary lists 'c-001' with a URL mismatch. Verify all .bib entries match the keys used in the text (e.g., ensure 'ray2026tauvoicebenchmarkingfullduplexvoice' exists in the .bib file with correct metadata).
- [writing] Complete the 'Limitations' section: The text cuts off mid-sentence in the Limitations section ('PCM-to- μ -law conversion introduces degradation...'). Ensure the full text is present and the section is logically concluded before the Ethics Statement.
- [writing] Verify figure compilation: Ensure all figure paths (e.g., 'figures/architecture/overview_v8.pdf') exist in the source directory and are correctly referenced. The PDF summary indicates no compiled PDF exists yet, suggesting build failures.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The paper demonstrates strong alignment between its quantitative claims and the data pre-

sented in Table 1 and the associated figures. The definitions of metrics like pass@1 , pass@k , and pass^k are consistent throughout the text and appendix. The reported performance gaps between S2S and cascade architectures are supported by the numerical evidence.

However, a significant issue exists in the bibliography. The entry ‘c-001’ references ‘https://github.com/goodfeli/dlbook_notation’, a repository unrelated to voice agent evaluation. This appears to be a placeholder or copy-paste error. If this citation is used in the text, it fails to support the attributed claim. Even if unused, its presence in the bibliography compromises the accuracy of the reference list.

Additionally, the introduction of the “Speech Fidelity” metric in Section 3.2 claims it is unique to the framework and uses an LALM-as-Judge. While the methodology is sound, the text does not immediately cite the specific judge model (Gemini 3 Flash) or the appendix section (E005) where the validation and model details are provided. To fully support the claim of novelty and methodological rigor at the point of introduction, the text should explicitly reference these details.

Finally, the summary in Section 4.2 regarding turn-taking scores (0.28–0.58 for cascade) is numerically consistent with Table 1 (min 0.283), but the rounding of the lower bound should be explicitly noted as an approximation to avoid any ambiguity regarding the exact minimum performance observed.

- [writing] Citation mismatch in bibliography: Entry ‘c-001’ points to ‘https://github.com/goodfeli/dlbook_notation’, which is unrelated to the paper’s content. This appears to be a placeholder error that must be corrected or removed to ensure reference accuracy.
- [writing] Section 3.2 introduces ‘Speech Fidelity’ as unique to EVA-Bench using an LALM-as-Judge but lacks an immediate inline citation or reference to the specific judge model (Gemini 3 Flash) detailed in the appendix. Add a reference to Appendix E005 or the specific model citation upon first mention to support the novelty claim.
- [writing] Section 4.2 states cascade turn-taking ranges from 0.28–0.58. Table 1 shows a minimum of 0.283. While rounding is acceptable, clarify if 0.28 is a rounded value or if the range excludes specific outliers to ensure precise alignment between text and data.

paper_reviewer_code_quality_paper

Verdict: minor_revision

The paper’s code quality artifacts, specifically the LaTeX source and implied repository structure, exhibit significant modularity and reproducibility issues that hinder evaluation from scratch.

First, the **modularity** of the appendix is critically low. The user simulator and agent prompts (Sections K and L) are consolidated into massive, monolithic text blocks within the LaTeX source. For instance, the ‘section_k_user_simulator_prompts.tex’ and ‘section_l_agent_prompts.tex’ files appear to contain thousands of lines of raw prompt text mixed with LaTeX formatting. This structure is prone to token truncation during compilation and makes version control or specific prompt iteration difficult. Following the principle

of splitting large files, these should be decomposed into a ‘prompts/’ directory with individual files per domain (e.g., ‘prompts/simulator/airline.yaml’, ‘prompts/agent/itsm.yaml’) and referenced via ‘\input’.

Second, **reproducibility** is compromised by missing dependencies. The source code relies heavily on ‘\input{tables/08_appendix/...}’ for critical data (e.g., model configurations, judge validation results). These table files are not present in the provided artifact list. Without these files, the paper cannot be compiled or the results verified. The repository must include all referenced ‘.tex’ and data files to satisfy the “reproducibility from scratch” requirement.

Finally, the **readability** of the metric definitions in ‘section_e_metric_details.tex’ is poor. The turn-taking scoring regime and latency breakpoints are embedded directly in the text without clear separation between the mathematical logic and the explanatory prose. This makes it difficult to extract the algorithm for implementation. A dedicated ‘metrics/’ module (either in code or a separate LaTeX file) with clear function-like definitions for the scoring regimes would improve clarity and maintainability.

While the experimental design is sound, the artifact organization requires refactoring to meet standard code quality expectations for a research framework.

- [writing] The paper’s code quality artifacts, specifically the LaTeX source and implied repository structure, exhibit significant modularity and reproducibility issues that hinder evaluation from scratch. First, the modularity of the appendix is critically low. The user simulator and agent prompts (Sections K and L) are consolidated into massive, monolithic text blocks within the LaTeX source. For instance, the section_k_user_simulator_prompts.tex and section_l_agent_prompts.tex files appear to contain th

paper_reviewer_data_quality_paper

Verdict: full_revision

The paper’s central claim relies on evaluating voice agents against “realistic” enterprise scenarios (Airline CSM, Healthcare HRSD, ITSM). However, a critical data quality failure is identified in **Section 5 (Limitations)** and **Appendix C (Data Generation Pipeline)**. The authors explicitly state: “Evaluation scenarios are fully synthetic; no real caller data or PII used” and that scenarios are generated by “SyGra (GPT-5.2 backbone).”

This constitutes **fabrication of input data**. The “ground truth” database states and “scenario databases” referenced in **Section 3.1** and **Appendix C.1** are not derived from real enterprise logs but are hallucinations generated by an LLM. Consequently, the “Task Completion” metric (Section 3.2), which relies on a SHA-256 hash of the ‘final_scenario_db’ vs. ‘expected_scenario_db’, is measuring the agent’s ability to satisfy an LLM-generated fiction, not a real-world constraint. The claim that the framework measures “realistic simulation” is unsupported when the underlying data is synthetic.

Furthermore, the **Reproducibility Statement** (Section 5) admits that “Full reproduction requires commercial model APIs” and that “results may vary by API version.” The

specific versions of the 12 evaluated systems (e.g., ‘gpt-5.4’, ‘gemini-3.1-flash-live-preview’) are not pinned in a machine-readable manifest, and the licenses for these proprietary APIs are not disclosed in the **Third-Party Dependency Licenses** section (Appendix E). This creates a “link rot” risk where the benchmark cannot be re-run or verified once API versions change or access is revoked.

The “scenario database” schema is also unvalidated. The generation pipeline (Appendix C.2) uses an LLM to create database states without a schema validator to ensure consistency with real-world constraints (e.g., valid date ranges, non-negative inventory). This introduces a high risk of “impossible states” in the ground truth, invalidating the “Task Completion” metric.

To proceed, the authors must either: (1) Replace the synthetic data with a subset of real, anonymized call logs and database states, or (2) explicitly reframe the paper as a “Synthetic Benchmark” and remove all claims of “realistic” or “real-world” fidelity. Additionally, a frozen manifest of all API versions and licenses must be provided.

- [fatal] The paper claims ‘realistic’ scenarios but admits in Section 5 that all 213 are ‘fully synthetic’ generated by LLMs. This is fabrication of input data for a benchmark claiming real-world task completion. Re-analyze using real logs or reframe as a synthetic benchmark.
- [science] External API dependencies lack version pinning or license details. The ‘Reproducibility Statement’ notes results vary by API version, yet specific model versions used for 12,780 trials are not verifiable. Provide a frozen manifest of all API versions and licenses.
- [science] The ‘scenario database’ schema is generated by LLMs without validation against real enterprise schemas. This risks impossible states (e.g., negative inventory), invalidating the ‘Task Completion’ metric. Validate the synthetic schema against a real DB schema.

paper_reviewer_figure_critic

Verdict: minor_revision

The figure suite in EVA-Bench is extensive and generally well-aligned with the paper’s quantitative claims, but several critical issues regarding legibility and self-containment must be addressed before publication.

Clarity and Legibility: The architecture overview (Figure 1, ‘figures/architecture/overview_v8.pdf’) is central to the methodology. In the current LaTeX source, the inclusion command does not specify resolution or vector properties. Ensure the final PDF is vector-based (not rasterized) to maintain legibility of the small text labels (e.g., “Simulator Validation”, “EVA-A”) when printed at standard conference size. Similarly, the perturbation analysis figures (Figures 3–7, ‘figures/perturbations/...’) rely heavily on color coding (accent, noise, both). The captions describe the colors, but the figures themselves must be checked for grayscale compatibility. If the bars for “accent” and “noise” are similar shades

of blue/green, they will be indistinguishable in black-and-white print. Consider adding distinct patterns (hatching) or ensuring high-contrast color palettes.

****Axis Labels and Units:**** Several figures, particularly the pipeline comparison (Figure 2, ‘figures/architecture_submetrics/fig_pipeline_comparison.pdf’) and the threshold sensitivity plot (Figure 8, ‘figures/threshold_sensitivity/threshold_sensitivity.pdf’), appear to rely on the surrounding text for axis definitions. The LaTeX source does not show explicit ‘\xlabel’ or ‘\ylabel’ commands for these plots. Every figure must be self-contained: the y-axis of the threshold sensitivity plot must explicitly state “EVA-X pass@1” and the x-axis “Turn-taking threshold (τ_{tt})”. Without these, the figure is ambiguous.

****Legend and Caption Consistency:**** In the perturbation figures (e.g., Figure 3, ‘paper_conversation_progression_perturbation_pooled_fig.pdf’), the caption uses LaTeX math mode to describe the legend keys ($\blacksquare$), but the actual figure likely uses colored bars. The caption should describe the *colors* used in the figure (e.g., “Blue: accent, Red: noise”) rather than abstract symbols, or the figure must include a visual legend. Currently, the caption’s description of “Colors: ...” is disconnected from the visual data if the legend is missing from the image itself.

****Data Density:**** Figure 11 (‘figures/trial_count/trial_count_grid_ci.pdf’) presents a grid of CI widths. With 12 systems and multiple metrics, the individual panels risk being too small to read. Verify that the font size in the final compilation is at least 8pt and that the grid lines do not obscure the data points.

****Alt Text:**** The LaTeX source lacks ‘\alttext’ or equivalent accessibility tags for these figures. While not always required for arXiv, they are essential for accessibility compliance. Ensure a descriptive alt-text is generated for the screen-reader version of the paper, summarizing the key trend (e.g., “S2S systems show higher turn-taking scores than cascade systems”).

****Conclusion:**** The figures effectively visualize the complex data but require a “print-ready” pass to ensure axis labels, color contrast, and legend clarity are robust.

- [writing] Figure 1 (architecture/overview_v8.pdf) and Figure 2 (arch-comparison) lack visible axis labels and units in the provided source context. Ensure all axes in the final PDF are explicitly labeled with metric names (e.g., ‘EVA-A Score’) and units (e.g., ‘0.0-1.0’) to meet print legibility standards.
- [writing] The perturbation figures (Figs 3-7) use square symbols ($\blacksquare$) for the legend keys in the caption text, but the actual bar colors are not defined in the LaTeX source provided. Verify that the color mapping (accent, noise, both) is distinct and printable in grayscale, and that the legend is embedded within the figure or clearly defined in the caption.
- [writing] Figure 8 (threshold_sensitivity.pdf) contains shaded bands for 95% CIs. Ensure these bands have sufficient contrast against the line plots at print scale (600dpi) and that the dashed vertical line is clearly distinguishable from the grid lines.
- [writing] The caption for Figure 11 (trial_count_grid_ci.pdf) references ‘one panel per metric’ but the figure file name suggests a grid. Verify the layout is not too dense for the

page width, causing axis labels to overlap or become illegible.

paper_reviewer_jargon_police

Verdict: full_revision

The paper suffers from significant jargon overuse and a lack of immediate definitions for specialized acronyms and custom macros, creating a barrier for non-specialist readers.

First, the manuscript relies heavily on custom LaTeX macros (e.g., `\framework`, `\passatone`, `\passatk`, `\passpowerk`, `\metricbehavior`, `\metricuserspeechfidelity`) that are never defined in the main body text. These appear as opaque symbols in the Introduction and Methodology sections. For instance, in Section 1, the text introduces `\framework` and metrics like `\passatone` without explaining what they stand for in plain English. A reader must cross-reference the Appendix or guess the meaning, which violates the principle of self-contained readability.

Second, several acronyms are used without definition at their first occurrence. "LALM" appears in Section 1 ("LALM-as-Judge") but is not defined until the Appendix. While "LLM" is standard, "LALM" is not, and the paper later mentions "AudioLLM" and "SpeechLLM" as synonyms, creating confusion. Similarly, "S2S" is used as a primary architectural category in the Abstract and Introduction but is only defined in the Appendix. "VAD" is used in Section 2 ("VAD or transcription receipt") without expansion.

Third, the metric naming convention is overly technical. Terms like $\text{pass}^{\{k\}}$ (reliability) and $\text{pass}@k$ (ceiling) are presented with mathematical notation that obscures their practical meaning. The text should explicitly state "reliability across k trials" or "probability of passing all k trials" alongside the symbol.

Finally, the phrase "bot-to-bot" is used frequently (e.g., Section 1, Section 2) without a brief parenthetical explanation that this refers to "simulated user-agent interactions," which might be clearer for a broader audience.

To improve accessibility, the authors must define every acronym and custom macro at its first use in the main text, replace opaque mathematical shorthand with plain English descriptions, and ensure that architectural terms like "S2S" are explained before being used as categories.

- [science] The paper suffers from significant jargon overuse and a lack of immediate definitions for specialized acronyms and custom macros, creating a barrier for non-specialist readers. First, the manuscript relies heavily on custom LaTeX macros (e.g., `\framework`, `\passatone`, `\passatk`, `\passpowerk`, `\metricbehavior`, `\metricuserspeechfidelity`) that are never defined in the main body text. These appear as opaque symbols in the Introduction and Methodology sections. For instance, in Section 1, the text intro

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a logically coherent framework for evaluating voice agents, but there are specific points where the textual claims do not perfectly align with the presented data or the experimental design logic.

First, the central claim in the Introduction and Conclusion that "no system exceeds 0.5 on both EVA-A and EVA-X pass@1" is technically correct based on the data (the highest EVA-A is 0.490), but the phrasing creates a logical tension with the immediate follow-up: "Only gptrealtime (0.47, 0.57) clears 0.4 on both." The shift from a 0.5 threshold to a 0.4 threshold in the same sentence weakens the impact of the "0.5" claim. If the 0.5 threshold is the critical failure mode, the text should emphasize that *no* system is robust enough to clear it, rather than pivoting to a 0.4 observation which implies the 0.5 bar might be arbitrary. The logic of the "Accuracy-Experience Frontier" would be stronger if the text explicitly defined the 0.5 threshold as a "production-ready" benchmark rather than just a statistical observation.

Second, the definition of the composite "pass" metric in Section 3.2 relies on a strict logical AND of component thresholds (Task Completion=1, Faithfulness $\geq$ 0.5, Speech Fidelity $\geq$ 0.95). The paper presents "Mean" scores for these components in Table 1 (e.g., Faithfulness Mean = 0.338 for the first system). The logical consistency of the "pass@1" metric depends entirely on the assumption that the "pass" rate is the intersection of the binary pass/fail outcomes of these components. The text does not explicitly state that the "pass@1" value is *not* an average of the component scores, which could lead to misinterpretation. Given that the Faithfulness Mean (0.338) is well below the 0.5 threshold, the logical implication is that the pass rate for Faithfulness is near zero, driving the composite pass rate to near zero. The paper should explicitly clarify that the composite score is a binary intersection rate, not a weighted average, to ensure the reader understands why a system with a 0.338 Faithfulness mean has a 0.207 EVA-A pass@1 (implying some trials passed despite the low mean, or the mean is skewed by failures).

Third, the claim in Section 4.1 that "judge stochasticity is minimal" relies on a variance decomposition analysis. However, the experimental design described in the Appendix (Section E) indicates that specific judge models (e.g., GPT-5.2, Claude Opus 4.6) were selected and fixed for each metric. If the "judge" variable in the variance model is not a random effect (i.e., they did not swap judges between trials to measure inter-judge variance), then the "judge stochasticity" being minimal is a result of the fixed design, not an empirical finding of judge stability. The logic of the claim "judge stochasticity is minimal" is only valid if the study included a randomization of judges or a multi-judge setup. If the study used a single judge per metric, the claim should be rephrased to "variance due to judge selection is not a factor in this study" or similar, rather than implying a measurement of judge reliability. The current phrasing suggests an empirical measurement of judge variance that may not have been performed.

- [writing] The paper presents a logically coherent framework for evaluating voice agents,

but there are specific points where the textual claims do not perfectly align with the presented data or the experimental design logic. First, the central claim in the Introduction and Conclusion that “no system exceeds 0.5 on both EVA-A and EVA-X pass@1” is technically correct based on the data (the highest EVA-A is 0.490), but the phrasing creates a logical tension with the immediate follow-up: “Only gptrealtime (0.

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several claims that extend beyond the specific constraints of the experimental setup, particularly regarding generalizability and the definition of composite metrics.

First, the Introduction and Conclusion assert that “no system exceeds 0.5 on both EVA-A and EVA-X pass@1.” However, the methodology defines a “pass” as a strict logical AND of all sub-metrics (e.g., Task Completion=1, Faithfulness $\geq$ 0.5, Speech Fidelity $\geq$ 0.95). Table 1 reports **mean** scores for these sub-metrics (e.g., Speech Fidelity is $\sim$ 0.98 for most systems), but it does not explicitly report the **composite** pass@1 rate derived from the strict thresholding logic. There is a risk that the authors are conflating high mean component scores with high composite pass rates. If the composite pass rate is indeed <0.5 , the paper must explicitly show the distribution of composite passes, not just the means of the components, to support the claim that the **framework** reveals a fundamental trade-off. Without this, the claim of a “0.5 ceiling” is an over-interpretation of the component means.

Second, the Conclusion generalizes findings to architectural classes: “S2S systems lead in experience... while cascade systems lead in accuracy.” This is an overreach given the sample size ($n=3$ S2S, $n=7$ Cascade) and the specific models chosen (e.g., GPT-Realtime vs. specific open-source cascades). The Limitations section correctly notes that “sample size precludes inferential claims about model classes.” The conclusion should be revised to state that **the evaluated S2S implementations** outperformed **the evaluated cascade implementations** in these specific domains, rather than implying a universal architectural law. The observed gap may be due to the specific prompt engineering or model versions used rather than the architecture itself.

Finally, the robustness analysis claims that “accent and noise perturbations expose robustness gaps.” The data supports this only for a single French accent and a single coffee-shop noise profile. The Limitations section admits “Experiments include only one accent... and one noise environment.” Extrapolating these results to general “accent” or “noise” robustness is unjustified. The text should be qualified to specify that these gaps were observed **under the tested conditions** (French accent, coffee shop noise) and may not generalize to other accents (e.g., Indian, Spanish) or noise types (e.g., street, office).

- [science] The claim that no system exceeds 0.5 on composite EVA-A/EVA-X pass@1 conflates mean component scores with strict AND-thresholded pass rates. Table 1 shows high mean Speech Fidelity ($\sim$ 0.98) but does not report the actual composite pass

rate. Justify the 0.5 ceiling claim with explicit composite pass statistics, not component means.

- [writing] The conclusion generalizes that "S2S systems lead in experience" and "cascade systems lead in accuracy" based on $n=3$ S2S and $n=7$ cascade systems. This overreaches the data, as the Limitations admit sample size precludes class-level inference. Soften to claim only that *evaluated* S2S implementations outperformed *evaluated* cascades in these specific domains.
- [writing] The robustness claim regarding "accent and noise" gaps extrapolates from a single French accent and one coffee-shop noise profile. The Limitations section admits this narrow scope. Qualify the findings as specific to the tested conditions rather than general acoustic robustness to avoid overreach.

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents a robust technical framework for evaluating voice agents but lacks sufficient depth in its safety and ethics disclosure, particularly regarding the potential for harm in the evaluated domains.

****1. Insufficient Harm Mitigation Disclosure (Sec. ethics, Sec. limitation):**** The authors explicitly state in the Limitations (Section 6) that there is "No assessment of harmful outputs or PII exposure." However, the evaluation domains include ****Healthcare HRSD**** (involving simulated medical credentials, leave requests, and sensitive employee data) and ****Airline CSM**** (involving financial transactions and identity verification). In a real-world deployment, failures in these areas could lead to privacy breaches, financial loss, or denial of critical services. The paper currently treats these as "task completion" metrics without addressing the *safety* of the agent's behavior. The Ethics Statement must be expanded to detail: * What specific content filtering or safety guardrails were active during the 12,780-trials? * How were "harmful outputs" (e.g., hallucinated medical advice, unauthorized data access) detected or prevented during the simulation? * A clear statement on whether the framework is intended to be used for safety-critical evaluation or purely for performance benchmarking.

****2. Adversarial Robustness and Simulator Safety (Appendix K):**** The User Simulator prompts (Appendix K) are designed to be "faithful" to the scenario but do not appear to include instructions for handling adversarial inputs or "jailbreak" attempts. If the Voice Agent is being evaluated for production readiness, the simulator should ideally include a "red-teaming" component or at least a discussion on how the framework handles inputs designed to elicit harmful behavior. The current setup assumes a cooperative user, which may mask safety vulnerabilities in the agent's policy adherence.

****3. Synthetic Data Privacy Risks (Sec. reproducibility):**** While the authors claim the data is synthetic, the Healthcare domain involves generating realistic-looking NPI numbers, DEA codes, and employee IDs. There is a non-zero risk that the generative process

could inadvertently reproduce real-world identifiers or patterns that could be reverse-engineered. The Reproducibility Statement should explicitly confirm that the released dataset has undergone a privacy audit to ensure no real PII leakage, even in synthetic form.

****Recommendation:**** The authors should add a "Safety and Risk Assessment" subsection to the Methodology or Ethics section. This should detail the safety filters applied, the nature of the "harmful output" risks considered (and why they were excluded or how they were mitigated), and a confirmation of the synthetic data's privacy safety. Without this, the paper's claim of being an "end-to-end framework" for enterprise domains is incomplete regarding safety.

- [writing] What specific content filtering or safety guardrails were active during the 12,780 trials?
 - [writing] How were "harmful outputs" (e.g., hallucinated medical advice, unauthorized data access) detected or prevented during the simulation?
 - [writing] A clear statement on whether the framework is intended to be used for safety-critical evaluation or purely for performance benchmarking.
2. Adversarial Robustness and Simulator Safety (Appendix K): The User Simulator prompts (Appendix K) are designed to be "faithful" to the scenario but do not appear to include instructions for handling adversarial inputs or "jailbreak" attempts. If the Voice Agent is being evaluated for production readiness, the simulator should ideally include a "red-teaming"

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence presented in EVA-Bench is robust in its scale (213 scenarios, 12 systems, ~12,780 trials) and the rigor of its metric definitions (e.g., deterministic task completion via SHA-256). The use of LLM-as-Judge with high inter-annotator agreement ($\kappa = 0.777\text{--}0.845$) provides a solid foundation for the subjective metrics (Faithfulness, Conciseness). The variance decomposition correctly identifies trial stochasticity as the primary noise source, justifying the multi-trial approach.

However, the evidence for the central claim regarding "robustness gaps" is weakened by the experimental design in the perturbation suite. The study tests only one specific accent (French) and one noise environment (coffee shop) across 90 scenarios (Section Limitations). While the results show significant drops for cascade systems, the sample size ($n=90$) and lack of diversity in acoustic conditions make it difficult to generalize these findings to "accent" or "noise" as broad categories. The observed effects could be artifacts of the specific French voice model or the specific spectral characteristics of the coffee shop noise used. To support the claim that "accent hits cascade accuracy" as a general architectural property, the authors must expand the perturbation suite to include at least two additional accents and two distinct noise types, or explicitly frame the current results as a proof-of-concept for the framework's capability rather than a definitive study of robustness.

Additionally, the handling of the 12.0% of trials that required regeneration due to simulator behavioral drift (Section 3.1) lacks statistical scrutiny. The paper does not report whether the regenerated trials exhibit different performance characteristics compared to the initial failures or if the regeneration process itself introduces a selection bias (e.g., only "easier" failures are successfully regenerated). A sensitivity analysis comparing metrics on the original vs. regenerated subsets is necessary to ensure the final reported scores are not inflated by this filtering process.

Finally, the calculation of the 'pass^k' metric assumes independent trials. Given that the simulation is bot-to-bot and relies on VAD and WebSocket timing, there is a risk of temporal correlation between trials (e.g., a specific VAD threshold causing consistent early cuts). The authors should verify the independence assumption or adjust the 'pass^k' calculation to account for potential intra-scenario correlation.

- [writing] The scientific evidence presented in EVA-Bench is robust in its scale (213 scenarios, 12 systems, ~12,780 trials) and the rigor of its metric definitions (e.g., deterministic task completion via SHA-256). The use of LLM-as-Judge with high inter-annotator agreement ($\kappa = 0.777\text{--}0.845$) provides a solid foundation for the subjective metrics (Faithfulness, Conciseness). The variance decomposition correctly identifies trial stochasticity as the primary noise source, justifying the multi-trial approach.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis in EVA-Bench is ambitious in scale but lacks necessary methodological transparency regarding inference. While the paper presents extensive data (12 systems, 213 scenarios, ~12k trials), the statistical rigor required to support the comparative claims is incomplete.

First, the handling of multiple comparisons is absent. The paper presents numerous pairwise comparisons between systems and architectures across clean and perturbed conditions (Figures 4-7). Asterisks indicating significance ($p < 0.05$, etc.) are used, but the specific statistical tests (e.g., paired t-test, Wilcoxon signed-rank, ANOVA) and, crucially, the correction method for multiple testing (e.g., Bonferroni, Benjamini-Hochberg) are not specified. With dozens of comparisons, the uncorrected false discovery rate is likely inflated, potentially invalidating the "significant degradation" claims.

Second, the construction of confidence intervals is ambiguous. Table 1 and Figure 3 report "95% percentile bootstrap intervals." However, the unit of resampling is unclear. The caption of Figure 3 mentions "systems-within-type," which implies resampling at the system level. For the Hybrid architecture ($n=2$ systems), a bootstrap CI is statistically unsound and likely overconfident. If the bootstrap is performed at the trial level within systems, the aggregation method to the architecture level needs explicit definition. The current description risks misrepresenting the uncertainty of the estimates.

Third, the variance decomposition claim in Section 4.2 ("Trial stochasticity is the domi-

nant variance source; judge stochasticity is minimal ($p < 0.0001$)” lacks the underlying model. A linear mixed-effects model or similar ANOVA framework is implied but not defined. Without specifying the fixed and random effects (e.g., System, Judge, Trial) and the resulting variance components, this claim cannot be verified.

Finally, the perturbation analysis (Section 4.3) reports mean deltas (Δ) and significance but omits the standard error or confidence intervals for these differences. Reporting only the mean delta without a measure of its variability makes it difficult to assess the reliability of the observed robustness gaps, especially given the small number of perturbation trials ($k=3$) compared to clean trials ($k=5$).

To ensure reproducibility and statistical validity, the authors must explicitly detail the statistical tests, correction methods for multiple comparisons, and the precise resampling strategy for confidence intervals.

- [writing] The statistical analysis in EVA-Bench is ambitious in scale but lacks necessary methodological transparency regarding inference. While the paper presents extensive data (12 systems, 213 scenarios, ~12k trials), the statistical rigor required to support the comparative claims is incomplete. First, the handling of multiple comparisons is absent. The paper presents numerous pairwise comparisons between systems and architectures across clean and perturbed conditions (Figures 4-7). Asterisks indicati

paper_reviewer_writing_quality

Verdict: minor_revision

The paper demonstrates a high level of technical depth, but the writing quality requires minor revisions to improve clarity, flow, and precision in specific sections.

Clarity and Precision: In Section 3.1, the statement regarding “12.0% of trials required regeneration” is slightly abrupt. The reader must infer which “four systems” were used for this specific validation metric, as the main experiment involves 12 systems. Explicitly stating “Across a subset of four systems...” or defining the scope immediately would prevent confusion. Similarly, in the Limitations section, the example regarding GPT-5.4 and GPT-5.2 judges is syntactically ambiguous. Rephrasing to clarify the version mismatch (e.g., “evaluated by a different model version”) would enhance readability.

Flow and Cohesion: Section 4.2 presents dense statistical comparisons. The sentence comparing S2S and Cascade turn-taking metrics (“mean turn-taking 0.82–0.83 vs. cascade 0.28–0.58”) is factually dense but slightly disjointed from the preceding sentence about the “Accuracy–Experience Frontier.” A transitional phrase linking the architectural dominance in experience to the specific metric values would improve the narrative flow. Additionally, the definition of the aggregate metrics (pass@1 , pass@k , pass^k) in Section 3.2 is rigorous but could be slightly more accessible. The sentence “A conversation passes a dimension only if all component metrics meet thresholds” is logically sound but could be smoothed to better integrate with the preceding bullet points.

Notation and Consistency: The paper relies heavily on custom LaTeX macros (e.g.,

\passatk, \passpowerk). While standard in technical papers, the text occasionally assumes the reader has the appendix open to decode these. In the main text, ensuring that the first mention of these metrics includes a brief, plain-English gloss (e.g., "reliability (pass^k)") would aid readability for a broader audience. The cross-referencing in the Appendix (e.g., "Table~\ref{tab:elevenlabs-params}") is generally correct, but the text sometimes jumps between referring to a table and describing its content without a clear visual anchor in the flow.

****Grammar and Syntax:**** The grammar is generally strong. However, there are minor instances of passive voice that could be tightened. For example, in Section 4.3, "Errors concentrate on alphanumeric entities" is clear, but "Errors are concentrated on..." or "We observe a concentration of errors on..." might offer a slightly more active voice. The phrase "under-representing disfluencies" in the Limitations section is correct but could be "under-represents" for better subject-verb agreement if the subject is "The simulator."

Overall, the paper is well-written but would benefit from these targeted clarifications to ensure the complex methodology is accessible without sacrificing precision.

- [writing] In Section 3.1 (Conversation Simulation), the phrase 'Across four systems, 12.0% of trials required regeneration' lacks a clear antecedent for 'four systems' and the specific criteria for 'regeneration' are not immediately defined in the sentence context. Clarify which systems were tested for this specific metric and define the regeneration trigger.
- [writing] In Section 3.2, the definition of 'pass@k' and 'pass^k' uses inconsistent notation in the text versus the Appendix. The main text uses \passatk and \passpowerk macros, but the explanation 'A conversation passes a dimension only if all component metrics meet thresholds' is slightly ambiguous regarding whether this applies to the aggregate or individual components. Ensure the logical flow of the pass criteria is explicit.
- [writing] In Section 4.2 (Main Findings), the sentence 'S2S systems dominate EVA-X (mean turn-taking 0.82–0.83 vs. cascade 0.28–0.58)' contains a range that is not immediately supported by the adjacent table values without calculation. Consider adding a brief parenthetical reference to the specific table rows or clarifying that these are aggregated means to improve readability.
- [writing] In the Limitations section, the phrase 'LLM/LALM judges may exhibit stylistic biases (e.g., GPT-5.4 evaluated with GPT-5.2 judges)' is confusing. It implies a version mismatch or a specific bias scenario that needs clearer phrasing, such as 'evaluated using a different model version (GPT-5.2) than the one generating the content (GPT-5.4)'.
- [writing] Throughout the Appendix (e.g., Section E, Metric Details), there are several instances where the text refers to 'Table X' or 'Figure Y' without the actual table/figure being immediately visible in the flow (e.g., 'Table~\ref{tab:elevenlabs-params}'). Ensure all cross-references are valid and the text flows logically even if the reader is not flipping back and forth.